

Types of Economic Behavior: The Instrument for Management of Individuals, Institutions, Countries and Humankind

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Abstract.

This is short preliminary paper. In next papers the detailed description of all aspects will be published.

The typology of economic behavior is constructed. The type of economic behavior is the mathematical (abstract) operator in the information space, which is constructed by a special method from the data about economical problem. It is shown that for description of any human behavior (his / her decision making) it is need only 16 two-component abstract information operators (2AIO), which are constructed by a special manner. It is shown that there are only 16 types of interaction (relationship) between 2AIO.

It is shown that typology of 2AIO can to describe wide field of human behavior. Authors carried out the identification of types 2AIO for more than 2 thousand persons from 1998 to 2011. *Ex post* and *ex ante* forecast was made for rational behavior for each person. The theoretical prediction for relationships between persons (as between types of 2AIO) was also placed (more than 3 thousands of forecasts). More than one thousand predictions have been done for the level of effectiveness for defined functional responsibilities of a person.

As examples there is the description: forecast the individual behavior (the causes for suicide of eminent physicist Paul Ehrenfest in 1933); dangers for the High Level Managers. A few theorems on development of mechanism design for application of game theory in economics are presented.

With using of 2AIO typology is an effective tool for solving such problems: the classification of the purposes for the person's behavior; the forecast of economic behavior for the person; the forecast of constraints for the person's behavior in a given interior of activity; the development of effective mechanisms and technologies for human resource management (e.g., motivation and incentives); the choice of the most effective trainings for the person, taking into account the interior of his/her behavior; the choice of optimal learning styles for the person.

The hierarchical systems for cooperative behavior (pyramids of behavior) from 2AIO were considered. It is shown that there are people who have the ability to introduce new information at a given level of the pyramid of behavior (or to adequately recover the missing it part at this level). These people are called *coordinators*. The coordinator is often named as “talent” or “genius”. The methods for identification of coordinators are constructed. It is shown that the pyramid of behavior with high-level (for example, at the country level) can be playing a role an institution. It is shown that if at least one player is the coordinator (for this problem), the results of the modern game theory are incorrect.

As examples there is the description: elite for the country; modern terrorism and National security (including the terroristic act September 11, 2001 in USA, Russia: 2002 and 2004, Spain: 2003, United Kingdom in 2007 and beyond; these technologies were forecasted in 1998 and presented in Internet in Russian); reasons of the financial crisis at 2008; three levels of loyalty (three-level of goals); democratic mechanisms are not applicable to the coordinators; the P versus NP Problem solving in the frame of 2AIO.

Obtained results can to change the life for billions people.

Keywords: economic behavior, joint economic behavior, typology, forecast, game theory, management, National security.

JEL classification codes: B41, C60, C70, C91, C92, C93, J24, M12, M54, O10, P11.

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Preamble

This literature review is not exhaustive. Its purpose is to give the main sources, in which there is a large base of reference. Most references reflect the personal preferences of the authors.

Rubinstein (2005) has noted that the main drawback of behavioral economic models is the lack of a clear and correct mathematical formalism. For example, there are the game theory and the theory of economic equilibrium in economy. Consequently, these papers described many real situations, but very few of these theoretical models, which can be applied to a wide range of situations and which have clear and well-defined borders of their application.

In our paper an instrument, which can be applied to the analysis of a wide range of real situations, is described.

Rubinstein (1998) has outlined the way for consideration of the restrictions, which are imposed on economic agents. He called it “bounded rationality”, which stressed that economic agents sometimes make irrational economic behavior. However, models that were considered in Rubinstein (1998), based on trying to “illogical presented in a logical form”. Therefore, attempts for each of the “typical” economic situations develop appropriate logical scheme would lead to too much “logic”.

Maybe it makes sense not to develop algorithms' “logic to irrational behavior”? Maybe it makes sense to develop algorithms for classification of economic situations? Our paper presents the variant of such a solution.

In Borghans et al (2011) is discussed and illustrates identification problems in personality psychology. It is shown that measures are determined by a few multiple traits and do not identify any particular trait unless incentives and other traits are controlled for. Consequently, the classification of personality traits should include the interior of the individual. Results show that using a set of unchanging traits of the individual (which does not depend on the context of his / her behavior) is not promising (Borghans et al, 2011).

But “individual characteristics” can be considered as elements of the typology of economic behavior (based on a standardized method of constructing a “model problem”). Our paper proposes a variant of realization of such a way.

In Heckman (2011) provides an overview of the possibilities, which are offered by modern psychology of personality to the economy. But stresses that personality psychology “works” mainly with the description of specific cases. Whereas the economy is now “works” mostly with models and their testing. In addition, personal

psychology mainly studied the correlation, whereas in the economy mainly studied the causal relationships. Almlund et al (2011) explores the power of personality traits both as predictors and as causes of academic and economic success, health, and criminal activity. There is also an extensive bibliography, too.

In our paper a typology of economic behavior in which conclusions are made causal relationships, is constructed.

It is shown in Card et al (2011) that “that the vast majority of field experiments (68%) are Descriptive studies that lack any explicit model; 18% are Single Model studies that test a single model-based hypothesis; 6% are Competing Models studies that test competing model-based hypotheses; and 8% are Parameter Estimation studies that estimate structural parameters in a completely specified model”.

This means that today the methodology of the economic field experiments still imperfect. In our paper we obtain of such a methodology.

In Jensen and Meckling (1994) discuss five alternative models of human behavior that are commonly used (though usually implicitly). It is shown that the “Resourceful, Evaluative, Maximizing Model best describes the systematically rational part of human behavior”.

In our paper we derived a new model for economic behavior (both for a person and a set of people), which has a high level of predictive power.

Introduction

The person’s meaningful (rational) behavior or decision-making could be described by the structure of Fig. A.

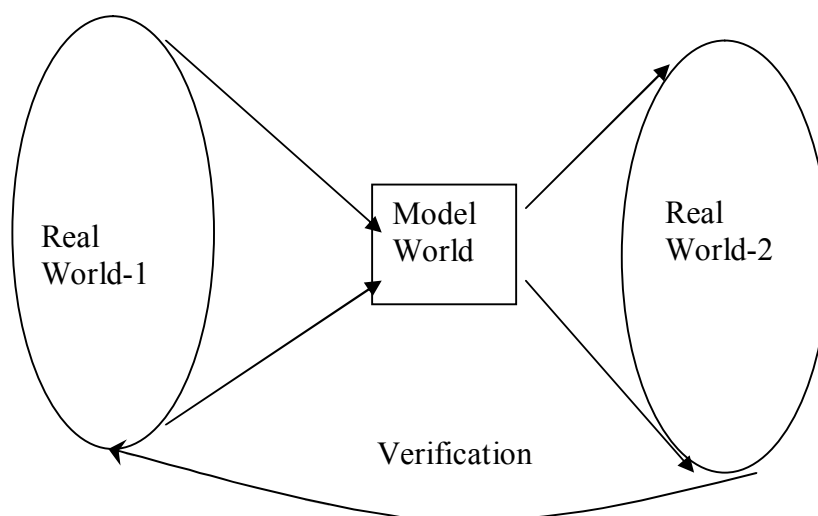


Fig. A

Below the words “intelligent” or “rational” will be missed.

If description of a person’s behavior (decision making) can be transferred to another person (to other people) without distortion – this mechanism is named as scientific knowledge.

How to describe the procedures for compressing and expanding data? How to take them into account for behavior modeling? Construction of instrument for description of the behavior of both the individual and the aggregate of people is the goal of this paper.

Note on the citation. If one refers to all previous authors, indicating that they have done, I will need more than one thousand pages. Therefore, references in the text will only on those works that describe the evidence and detailed reasoning of certain aspects.

1. The Information Space Constructing

Nature has hierarchical construction. Therefore, the mechanism from Fig. A should be *the same* for each of hierarchical levels. The differences can be appeared only from its use.

The every hierarchical level can be described by such categories of data (regardless of its origin):

- 1) about the objects, from which it consist (and which are “identical” in some sense);
- 2) about the interactions between such objects (owing to they come together into a coherent whole – a hierarchical level);
- 3) about the border (boundary) of the hierarchical level (which separates this level from the rest world);
- 4) about the structure of the hierarchical level (e.g., about the *selected* objects and/or connections between them, the violation of which leads to *destruction* of the level).

Finally, it is necessary for each of the above categories of data to describe their static and dynamic characteristics (i.e., their constancy and variability).

Total we get the 8 classes of data (classes of information) that constitute the *Information Space* for the problem of performing the behavior and / or decision-making. These 8 classes of information (*information components*) are the *basis* (in mathematical sense) for it.

Note. Any system and any database one can be similarly described.

The method for constructing of this *Information Space* is given in Table 1.

Table 1. The method for constructing of the Information Space.

Data about event	data about a class of similar events (the <i>generalizing</i> information components)	supporting (basic) elements for a class (<i>structure</i> , topology)	State, invariability	St-S
			Dynamics, variability	St-D
		<i>boundary</i> between this class and other	State, invariability	Bn-S
			Dynamics, variability	Bn-D
	data about this single unique event (the <i>detailing</i> information components)	the event as an single and unique (<i>object</i>)	State, invariability	Ob-S
			Dynamics, variability	Ob-D
		<i>connections</i> with other the events, which are like it	State, invariability	Con-S
			Dynamics, variability	Con-D

Thus, the mechanism of data compression involves the transition from Real-World objects to the objects of the Model World by way of the Information Space constructing, which meets of the requirements behavior and / or decision making.

If you agree with the above arguments, then *you have to agree* with the presented in sections 2 and 3 results, as they are simply the logical conclusion from the above results.

2. Operators in the Information Space

The human behavior or decision-making, as described in Fig. A, requires *the comparison* of two Real Worlds: one *before* and one *after* of the act of behavior (decision making). Thus, a person *before* the implementation of the behavior (or decision) must to carry out a comparison of two Real Worlds, *before* and *after* his act. But such a comparison he can *rationally* to make only in the information space (IP).

Thus, the human behavior (or his / her decision making) can be described as the operator, which translates IP up behavior of I_{before} (“desired” PI) into I_{after} – IP after the act of behavior (“received as a result of behavior” IP).

Definition. The object that receives the specific content of IP components and which transforms them, one is named an *abstract information operator* (AIO).

Thus, the AIO has input – output structure, which is typical for description of the black box. The unit block *accepts* (identifies, perceives) of fullness of the components of the Information Space I_{before} , and output block *changes* (creates, transforms) components of I_{before} , thereby forming the Information Space I_{after} .

It is easy to prove such theorems.

Theorem 1. For description of any human behavior (his / her decision making) it is need only such kind of AIO, who perceives only one component of information (from I_{before}), and who changes also only one component of information (from I_{after}).

Sketch of proof. The activity result for any AIO's can be represented as the activity result of the set of abstract information operators, which has only one input information component and has only one output information component (we denote it as g_{ik}). Note that every g_{ik} is a separate black box (of course, we must enter a sequence of Information Spaces $I_{before} \rightarrow I_{after}^1 \rightarrow I_{after}^2 \rightarrow \dots \rightarrow I_{after}$ for each such g_{ik}). $\square$

Theorem 2. The operator g_k^i is commutative $g_i^k + g_n^m = g_n^m + g_i^k$ and associative $g_1 + (g_2 + g_3) = (g_1 + g_2) + g_3$.

Theorem 3. Total number of operators g_k^i have 64 different versions.

Proof. The input block may to contain one from 8 information components from I_{before} , and output block may to contain one from 8 information components from I_{after} . $8 \times 8 = 64$. $\square$

Let we consider such class operators from g_k^i .

Definition. AIO is named as a two-component (2AIO) when it satisfies the following conditions (components and their classes are shown in Table 1.):

1. It in the input block has only one information component and in the output block – too;

2. It has one information component as a *dynamic*, and another – as a *static*.

3. It has one information component as a *generalizing*, another – as a *detailing*.

Background and comments see Shiyan (2011a).

Theorem 4. For description of the every behavior (or decision making) 16 types 2AIO are enough. There are these types of 2AIO:

$$\begin{aligned} &\langle St-S | Con-D \rangle, \langle St-S | Ob-D \rangle, \langle St-D | Con-S \rangle, \langle St-D | Ob-S \rangle, \\ &\langle Bn-S | Con-D \rangle, \langle Bn-S | Ob-D \rangle, \langle Bn-D | Con-S \rangle, \langle Bn-D | Ob-S \rangle, \\ &\langle Ob-S | St-D \rangle, \langle Ob-S | Bn-D \rangle, \langle Ob-D | St-S \rangle, \langle Ob-D | Bn-S \rangle, \\ &\langle Con-S | St-D \rangle, \langle Con-S | Bn-D \rangle, \langle Con-D | St-S \rangle, \langle Con-D | Bn-S \rangle. \end{aligned}$$

Thus, the classification for *all possible types* of behavior and decision making is made by Theorem 4.

For identifying the type of 2AIO (that is for allocating a specific type from the set of all 16 types of 2AIO) have the following theorem.

Theorem 5. For identification of the 2AIO type it is necessary and enough to use four different dichotomous procedures.

Proof. $2^4=16$.□

We describe an example of such a division (the dichotomies can be used in *arbitrary order*: the result is independent of the sequence).

The input block of 2AIO is described with using of such two dichotomies.

The first dichotomy. If input block has a generalizing information component – that 2AIO is named as *generalizing* type. In the opposite case – 2AIO is named as *detailing* type (and then the input block has a detailing information component).

Thus, the generalizing and detailing types of 2AIO making a kind of “inverse” behavior (accept of “inverse decision making”) – see Fig. B.

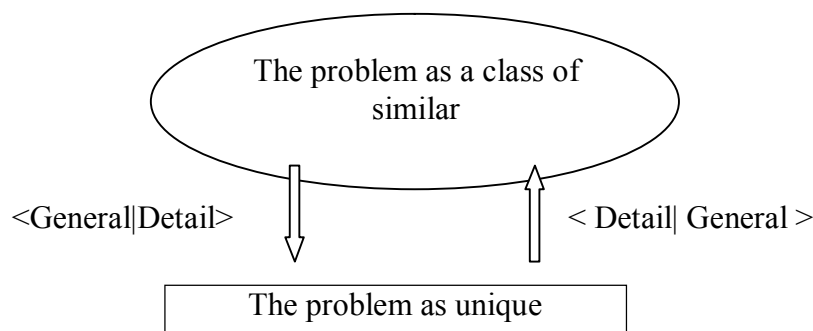


Fig. B

The second dichotomy. If input block has a dynamic information component, then 2AIO is named as *participant* type. In the opposite case (input block has a static information component) – 2AIO is named as *observer* type.

The *presence* in 2AIO of a class of information components is described by third and fourth dichotomies (irrespective of their location on the input or output blocks).

The third dichotomy. If 2AIO has objects information components – then it is named as *object-oriented* type. In the opposite case (2AIO has connections information components) it is named as *connection-oriented* type.

The fourth dichotomy. If 2AIO has structure information components – then it is named as *uniting* type. In the opposite case (2AIO has boundary information components) it is named as *demarcating* type.

Theorem 6. Each 2AIO is uniquely specified by a set of their 4 dichotomies and can be represented as a vector $\{a,b,c,d\}$, where a, b, c, d take the values 0 or 1.

Note. The concrete value of the dichotomy pole by matching 0 or 1 for further discussion does not matter (it corresponds to just fix one of 2AIO as a “reference point”).

Definition. The set of 2AIO (vectors $\{a,b,c,d\}$) is denoted as $\{T_{ij}\}$, $i=1, \dots, 16$.

3. Typology for interaction between types of 2AIO

We introduce on the set $\{T_{ij}\}$ the class operators $\{e_{ij}\}$, which transform one type 2AIO into another type of 2AIO. The representation of this operator is a 4-component vector, which is added to the vector type 2AIO. Vector components have values of 0 or 1, and addition we define from *mod 2*.

Basis in $\{e_{ij}\}$ are 4 vectors: $e_1=\{1,0,0,0\}$, $e_2=\{0,1,0,0\}$, $e_3=\{0,0,1,0\}$ and $e_4=\{0,0,0,1\}$ (these vectors are dichotomies introduced by Theorem 6). Thus, knowing the description of the dichotomies, we can make a description of all the operators of $\{e_{ij}\}$. Thus, the operators from $\{e_{ij}\}$ can be described using the parameters from behavior and decision making.

Definition. We name the vector from $\{e_{ij}\}$ as relationship between the relevant operators 2AIO.

It is easy to see that there are only 16 different operators, which operate on the set $\{T_{ij}\}$. There are such operators in addition to the above four basic operators and the identity operator $e_0=\{0,0,0,0\}$: $e_5=e_1+e_2$, $e_6=e_1+e_3$, $e_7=e_1+e_4$, $e_8=e_2+e_3$, $e_9=e_2+e_4$, $e_{10}=e_3+e_4$, $e_{11}=e_1+e_2+e_3$, $e_{12}=e_1+e_2+e_4$, $e_{13}=e_1+e_3+e_4$, $e_{14}=e_2+e_3+e_4$ и $e_{15}=e_1+e_2+e_3+e_4$.

Note. The pole of dichotomy “generalizing – detailing” in vector of 2AIO type must to change after the action of the operator e_1 . Therefore, the second and third components in a vector of 2AIO must to exchange after e_1 action. As a result, the law of action for operators e_5-e_{15} we define as follows: first always do the operator e_1 (resulting in second and third component of 2AIO have reversed by places), and only then do the other operators e_2-e_4 .

Such theorems give the mathematical structure of the set $\{e_{ij}\}$.

Theorem 7. A set of operators $\{e_{ij}\}$ is noncommutative group (e.g., $e_7 \bullet e_{13} \neq e_{13} \bullet e_7$).

Theorem 8. Group $\{e_{ij}\}$ consists from three sub-groups. The first consists of a single operator (“identical relationship”), which has the property: $e_0 \bullet e_0 = e_0$. The second consists of 11 operators (“symmetrical relationships”), which has the property

$e_i \bullet e_i = e_i^2 = e_0$. The third consists of four operators (“asymmetrical relationships”), which has the property $e_i^4 = e_0$.

Theorem 9. Each of asymmetric operators divides a set $\{T_{ij}\}$ into four nonintersecting orbits, which are consisted from four different types of 2AIO.

Corollary. Set $\{T_{ij}\}$ is a direct sum of four sets, each of which was created by the operator of $\{e_{ij}\}$, which has the property $e_i^4 = e_0$.

Asymmetric relationship e_{13} is selected, because it only provides the highest level of self-programming in a set of 2AIO. In fact: only under the action e_{13} : $2AIO_1 \rightarrow 2AIO_2$ the input block of $2AIO_2$ is the same as the output block of $2AIO_1$. Owing to $2AIO_2$ “is programmed” by behavior (decision) of $2AIO_1$. The orbit for operator e_{13} we named as “ring of self-programming”.

The set of operators $\{e_{ij}\}$ can be represented in the form of graphs (which connect two points – two 2AIO). The asymmetrical relationships are directed graphs.

The method for combining 2AIO for joint behavior (decision) is given by a theorem.

Theorem 10. For each of 2AIO from $\{T_{ij}\}$ has only one 2AIO, when the *joint* behavior (decision) both types will be most effective. This type of 2AIO is given by relationship e_{14} .

Sketch of proof. Two 2AIO, which are related by relationship e_{14} , can not carry out of effective control: each of them operates with characteristics that have been excluded for the other (see Table 1 and Fig. 2). These types in input block “see” fundamentally different characteristics, and therefore they will not “understand each other”. Relations e_2 , e_3 and e_4 are quite acceptable for a pair of types: they use “familial” classes of information components, which are *complement* each other (into generalizing or detailing classes). Thus, the required pair of types 2AIO can be connected by the relationship e_{14} . □

Let we construct such structures in $\{T_{ij}\}$.

Let we select an arbitrary 2AIO1. With using operator e_{13} we construct the orbit (named as “ring of self-programming”). Then we find, using relationship e_{14} , another $2AIO_2$ («joined» for $2AIO_1$). For $2AIO_2$ we construct the orbit too. Now we *connect* the two orbits so that operators $2AIO_1$ and $2AIO_2$ were in “node” of such dual orbit (named as “ring joined self-programming”). Thus in each node two 2AIO are jointly connected by relationship e_{14} too. Thus, 8 different 2AIO are already joined into one structure.

The remaining of 8 2AIO from $\{T_{ij}\}$ can be similarly joined into second ring of joined self-programming.

Finally, the set $\{T_{ij}\}$ is a direct sum of two different rings of joined self-programming, which are constructed as above.

Can “to join” all 2AIO from $\{T_{ij}\}$ into single structure? The answer is “yes”.

Let we consider an arbitrary 2AIO and his ring of joined self-programming. The second ring of joined self-programming can be added to our 2AIO only with two different ways: so that a single “node” was only such 2AIO that have the same pole for dichotomy “detailing – generalizing” as our 2AIO.

Thus, the structure is constructed from all the 16 elements of the set $\{T_{ij}\}$, where each node is 2AIO brings together six orbits. This structure has nontrivial topological architecture.

Let we consider an arbitrary orbit in $\{T_{ij}\}$, created by relationship e_{13} , which is represented as a graph. This orbit can be considered as two-dimensional manifold, where the graph orientation e_{13} induces the orientation of the edges. Now let we glue the four orbits (obtained 4 different manifolds) in the manner described above. As result the two-dimensional surface will be obtained with a nontrivial topological structure: it is diffeomorphic to 2D-sphere with seven glued Mebius films. It is interesting that a bunch (bouquet) of 6 circles have exactly the same topological properties.

We emphasize that in described above construction the relationships e_{13} and e_{14} between 2AIO are fixed. That is 2AIO or transfers its behavior (as a baton) to another 2AIO by the graph e_{13} , or both carries out the behavior in the joined pair 2AIO (which are connected by relationship e_{14}).

Definition. The described above construction on the set $\{T_{ij}\}$, which includes all 16 of its elements (2AIO) and which contains all 16 operators from $\{e_{ij}\}$, is named as *socion*.

Thus, socion is the object which must be formed from 2AIO for implementing a maximum variety of possible behavior (for taken all possible types of behavior) or doing the most of the analyzed choice decision.

The case is implemented in socion, when one 2AIO provides effective communication with the greatest possible number of other 2AIO.

The instrument is described, and we can to use it.

4. A person as 2AIO

There is a set of theorems which are consequences above.

Theorem 11. Every 2AIO uses only 25% of the information.

Sketch of proof. $2/8=0.25$. $\square$

Corollary. Separate 2AIO is not able to implement effective behavior in all situations.

Theorem 12. Separate 2AIO is capable of functioning in the socion only and only when he has the ability to *simultaneously* communicate with 7 ± 2 other types 2AIO.

Sketch of proof. Socion is a bouquet of six circles, which are connected with all types of 2AIO. So everyone 2AIO should have the ability to connect five other rings (with links to 5 other types). One obtains the maximum number of connections, when to add 4 types, which are associated with asymmetrical relationship with our 2AIO. $\square$

Theorems 11 and 12 probably are the first theoretical proof of the known properties of persons: 1) near 20-25% of new information remain in long-term memory, and 2) the “magic Miller number” 7 ± 2 . Thus, 2AIO is good tool for description of general patterns for person’s behavior.

Proposition 1. Every 2AIO is the type of *rational* behavior of a person.

Grounds are given in Shiyan (2011a), which describes the general method for type identification.

The word of the “rational” is used in sense of Fig. A, - when a person does the choice.

Note. 2AIO do not describe a type of person’s rational behavior as the whole. There is the *normative* behavior, when a person acts into the frame of the social and economical norms. This behavior does not require the choice.

Authors carried out the identification of types 2AIO for more than 2 thousand persons from 1998 to 2011. *Ex post* and *ex ante* forecast was made for rational behavior for each person. The theoretical prediction for relationships between persons (as between types of 2AIO) was also placed (more than 3 thousands of forecasts). More than one thousand predictions have been done for the level of effectiveness for defined functional responsibilities of a person.

All forecasts were confirmed in full.

Note. Many people have studied the method of type 2AIO identification and forecast technology of human behavior from the books of authors (see Shiyan (2011a)). Students in the case-study did so as well.

We chose the public people as examples (politicians, scientists, etc.).

Example 1. Forecast the individual behavior. The causes for suicide of eminent physicist Paul Ehrenfest in 1933 (for details see Shiyan (2011)).

Professor Paul Ehrenfest from Leiden University was head of successful school of physics. Long-term depression was preceded by suicide in 1933.

Albert Einstein and Niels Bohr very appreciated of P. Ehrenfest, and were his *big friends*.

Types of 2AIO for participants are: Ehrenfest – $\langle \text{Con-S} | \text{St-D} \rangle$, Albert Einstein and Niels Bohr have the same type – $\langle \text{St-D} | \text{Ob-S} \rangle$. 2AIO types were identified from the description of biographies.

The type $\langle \text{Con-S} | \text{St-D} \rangle$ behavior in the interior of theoretical physics will be briefly description.

It type independently creates new logical chains with extremely difficult: it always relies either on a “general authority” or on previously learned material. It is the *detailing* type, and tends to rely “on the old”. It can not see “the entire mosaic of Physics” at once (it sees only the fragments). Therefore, it preferred to operate in constant conditions, but not when the physics is in the “explosion” (the period of quantum mechanics). Moreover, this “explosion” is accompanied by a need to study the mass of the new mathematical formalism. This type of very slowly accustomed to the new “rules of the game”.

Ehrenfest needed *help* from *other physicists* to understand the “new physics”. His *best friends* Einstein and Bohr could help for him! But the *new information* between these types of people goes in *one direction only*: $\langle \text{Con-S} | \text{St-D} \rangle \rightarrow \langle \text{St-D} | \text{Ob-S} \rangle$. That is, in the direction from Ehrenfest to Einstein and Bohr. But it is necessary in this case, *vice versa*. Moreover, Ehrenfest perceives as a direct criticism of his own results the explanation of new material which did these two physicists.

Therefore, a strong stress was formed in Ehrenfest to the top of the 1930s, which was given its type 2AIO. It stress was caused by his *type* difficulties with the development of “new physics”.

Ehrenfest is perceived by Einstein and Bohr as very “important” and a *clever* physics: He describes the *new* just in such way as it these physics perceive. Therefore, Einstein and Bohr emphasized demonstrate goodwill and friendship to Ehrenfest.

However, Ehrenfest thinks that this attitude is the appearance of *pity* to him. He surprised that his friends are “instantaneously” to understand the results that he did a *very long time*.

To Ehrenfest is important the ability to “serve the people” and to be a *key member* of the “great movement”. The tendency to rise as high as possible in the hierarchy is characterized to him. But Ehrenfest “saw” that the physical society of his “rejects”: it He “saw” owing to communication with their friends.

The suicide is a result...

This example shows an amplification mechanism of stress owing to communication of certain 2AIO types each of other.

It is interesting that to *rectify* the situation would be quite easy: it had to ensure communication with Ehrenfest physicists that would *suit* him 2AIO types. In detail see Shiyan (2011a). □

Example 2. It is shown in Shiyan (2011a) that the 2AIO typology is an effective tool for solving such problems:

1. The classification of the purposes for the person's behavior;
2. The forecast of economic behavior for the person;
3. The forecast of constraints for the person's behavior in a given interior of activity;
4. The development of effective mechanisms and technologies for human resource management (e.g., motivation and incentives);
5. The choice of the most effective trainings for the person, taking into account the interior of his/her behavior;
6. The choice of optimal learning styles for the person.

Note. The constraints of specialty for the person are given by a set of interiors, where he/she performs the behavior.

In detail will be described in separate paper. □

Example 3. Dangers for the High Level Managers.

There are the special constraints on strategies for each 2AIO type. In general case, the utility functions of economic agent should be written in form: $U(x_i)$, where $i=1...16$, $x_i \in X_i \subset X$. X is the space of all possible strategies for agents. In this case $X_i \cap X_j \neq \emptyset$ for $i \neq j$.

This means that the alone person is not capable of adequate behavior and choice-making (except a set of specific interiors of behaviors that fit his/her 2AIO type). He/she cannot to do the *effective management*.

Moreover, an adequate transfer of information (result of analysis) can do only between identical 2AIO types. The *distortion* of information is in all other cases: one can to describe it with using of 2AIO tools (and today it is still *unknown* for managers).

Thus, it is necessary to organize a *special support* for senior management. For more details see Shiyan (2011a) and will be described in separate paper. □

Example 4. Economic behavior and game theory.

There are a number of theorems that *complement* the existing results of game theory. They are necessary for the development of mechanism design for application of game theory in economics. A few examples are below.

Theorem 13. It is needed to choose one of 16 sets of different strategies for each player for constructing of the mechanism design with using of game theory.

Theorem 14. There is a situation in mechanism design today, when a player do not know his type of economic behavior (his 2AIO type).

Consequence. Today, the type of player to 2AIO is unknown for himself / herself and for other players. Owing to there is information asymmetry. Thus, all results, which are now available in game theory, when applied to economics, must be only considered in the framework of information asymmetry.

Theorem 15. The maximum possible broad coalition of players in the coalition games consists of 16 2AIO types who jointly realized an economic behavior in the socion.

In detail will be described in separate paper. □

5. Hierarchical systems for cooperative behavior

There is a basic condition for the formation of hierarchical structures for realization of behavior (pyramids of behavior): the behavior at each hierarchical level is carried out by using people who have 2AIO types.

Proposition 2. There are people who have the ability to introduce new information at a given level of the pyramid of behavior (or to adequately recover the missing it part at this level).

Definition. People from Proposition 2 are called *coordinators*.

Note. The coordinator is often named as “talent” or “genius”. The methods for identification of coordinators are described in Shiyan (2011a and 2011b).

Pyramids of behavior are capable of functioning only and only in the presence of coordinators.

Definition. People which are no-coordinators are called *replicators*. They can to do only with using of well-known norms of behavior, communication and action at this level.

Replicators are the “ordinary” people.

Are there coordinators? Answer: Yes. For example, it is clear in interior of science, economics, management, art, etc.

Construction of a pyramid of behavior is made owing to such iterative algorithm: the *new behavior* is transmitted at a level down owing to activity of

coordinator. There the *new behavior* is transformed into all possible strategies at this level owing to activity of socion.

Table 2 shows the number of people in the pyramids of behavior depending on the number of levels (levels are measured from the bottom, from replicators). It is assumed that only one coordinator with corresponding level is “above” in the pyramid of behavior. The number of people in pyramid of behavior is *minimal*. In general, their quantitative composition is several times more.

Table 2.

The number of levels in the pyramid of behavior	The number of people in the pyramid of behavior
0	1
1	17
2	273
3	4,369
4	69,905
5	1,118,481
6	17,895,697
7	286,331,153
8	4,581,298,449

Theorem 16. Pyramid of behavior can effectively function only and only in the case, if all the places in it were “busy” (by both coordinators and replicators).

Sketch of proof. If in the pyramid of behavior at least one place is not a filled, then the corresponding behavior can not be done, and the pyramid of behavior will not be effective. □

Corollary. Each person has “equal importance” into pyramid of behavior.

Theorem 17. Pyramid of behavior exists a limited time: as soon as she finished the appropriate behavior, she begins to disintegrate.

Sketch of proof. In accordance with Fig. A *every* behavior has a purpose. Once the goal is reached, the continuation of *the same* behavior is meaningless. □

Corollary. In economics, the law should create the conditions to ensure that firms were able to *quickly* and *easily* to create and to disintegrate. In this case fragments of firms (i.e., the pyramid of behavior of lower level) may in future be included in the new firm as part of them.

Theorem 18. The person is the coordinator only in relation to the now existing pyramid of behavior. For *other* pyramids of behavior he may act in them as replicator.

Sketch of proof. The concept of “coordinator” is defined only in relation to the implementation of definite behaviors. □

Theorem 19. The pyramid of behavior with high-level (for example, at the country level) can be playing a role an institution.

Sketch of proof. The institution is a set of norms and rules (“rules of game”) for a certain group of people. The institution is shown at the state level. It acts as a kind of group of people who *together* decide a particular problem. But this is a pyramid of behavior. □

Theorem 20. If at least one player is the coordinator (for this problem), the results obtained in the modern theory of noncooperative games, for a given problem is not applicable.

Sketch of proof. Coordinator *changes* the rules of the game (as the term is defined). But today in game theory *is not a tool* to take this into consideration. Consequently, *new* game formulations and *new* game concepts of solutions theory are need to develop, which should take into account the *special role* of coordinator. □

Theorem 21. If at least one player is the coordinator (for this problem), the results of the modern theory of cooperative games are incorrect.

Sketch of proof. Hierarchically organized structures are not considered within the framework of cooperative games. □

Example 5. Elite for the country.

As shown in Table 2, most countries can implement behavior, which correspond to the five-level pyramid. Only a few countries (e.g., USA, Russia, China or Japan) can correspond to the behavior of six-level pyramid.

Interestingly, the EU will be able to construct a six-level pyramid of behavior only when the *unitary management* will be took place in EU. Today EU is a set of five-level pyramids of behavior, and therefore *loses* to countries, in which can be six-level pyramid (above all - the U.S.).

There are no more than 4.5-5 thousands of people in the *three highest levels* of the pyramid of behavior for each of the developed countries. This *small number* of people develops and implements the behavior of country *as a whole*. These people are called the *elite*. If the country has created conditions for: 1) selection, 2) education, 3) training and 4) activity for coordinators with appropriate levels, the behavior of this country will be close to optimal.

It can be shown that a part of the conditions, which are necessary for “success” of higher levels coordinators, had already been implemented in developed countries. Other conditions, which have not been made, are described in Shiyan (2011a). □

Example 6. Modern terrorism and National security.

Coordinator with higher level can be able to combine several different pyramids of behavior with lower levels. In developed countries the *narrow specialization* has traditionally encouraged (including for the coordinators too). As result, there are *new channels* for the National security of countries.

For example, today the media institutions are actively used for destabilization the social and economic status in the developed countries. The destabilization can be more effectively achieved in the cases, when mass media institutions are more developed. The reason is that modern terrorism is aimed at serving to media by *newsworthy*. In fact, modern terrorist acts are for the media as ready “*pictures*”.

Modern terrorists are not put forward any claims or explanations of his behavior. As a result, the media have *long-term newsworthy*. And therefore they attract a large number of different *experts* for maintenance of interest to him. As a result, the citizens will be “frightened” for a long time owing to activity of very qualified professionals.

In fact, control on media is intercepted by the modern terrorist technologies. Journalists (even coordinators!) are *narrow specialists*: they are not able to predict the consequences of their activity *for the country*.

To counter of modern terrorism it is necessary *to transform the way of operation* for a number of institutions in developed countries. Details and mechanism design are described in Shiyan (2011b).

Note. This channel of instability has been detected and described by one of the authors (AS) in 1998. Technologies for resistance have been developed. In 1998, the embassies of several developed countries have been informed. There is no answer. The fall of 1999 the description of channel instability and technologies of resistance were put on the Internet. Since September 11, 2001 these technologies have been used many times (Russia: 2002 and 2004, Spain: 2003, United Kingdom in 2007 and beyond, etc.). Technologies of resistance to modern terrorism had not been used. In detail see Shiyan (2011b) and will be described in separate paper. □

Example 7. Reasons of the financial crisis at 2008.

Financial institutions are *stand out*: they require high-level coordinators (because they manage the behavior of large numbers of people). In addition, each

coordinator has 2AIO type. There are therefore such threats to the economy from financial institutions:

- 1) The top management is not the coordinators with the necessary high levels;
- 2) Senior managers can make *different decisions in the same situations* (depending on the 2AIO type, sometimes even opposite).

The 2008 financial crisis had been caused by a no optimal choice of the strategies in financial activity by *a number* of leading financial organizations in the USA. This happened because their top managers were the coordinators with not appropriate levels.

There was an inefficient incentive system for decision-making managers. This has caused the lack of coordinators with necessary level and the difference in 2AIO types for senior management, too.

In detail will be described in separate paper. □

Example 8. Three levels of loyalty (three-level of goals).

Three levels of hierarchy for loyalty can be identified from Table 2 for each person. The first level is the loyalty to himself (level 0 in Tab. 2). The second level is the loyalty to a group of people, company, institution, the reference group etc. (2 - 4 levels in the Table 2). The third level is the loyalty to ethnic group, country, religion, ideology or to humanity at large (5 - 7 levels in the Table 2).

For the purposes, which have a person, one can to identify the same three-level hierarchy. Coordinator formulates the goal of behavior for the lower levels of hierarchy.

There are now several countries, in which goals of level 3 have been set (e.g., U.S., Russia, and China).

For the EU the behavior goal is absent. Moreover, the number of pyramids of behavior is increasing in EU. They parallel exist, without cooperating with each other. For example, there are the pyramids of behavior, which are formed by ethnic or religious criterions. The riots in France and several European countries (for example, in 2005) had this cause. Riots in the UK in 2011 can be as consequence in lack of goals at level 3 (and for young people – at level 2).

Interestingly, the *Country-level goals* are at a lower level than the "common to all Mankind". For example, people had been "polarized" by the *Communist* goal much more strongly than by the goals of *Nazism* or of *Patriotism*. One example (at a Country level) is comparison the time of existence of Communist regime in USSR and Nazism regime in Germany. Well known spy are another example at a people level: a physicist Professor Klaus Fuchs, the Cambridge spy ring (Kim Philby,

Donald Maclean, Guy Burgess, Anthony Blunt and, probably, John Cairncross), as well as famous writers of the twentieth century.

The goals of level 3 can deliver only a very small number of coordinators (see Table 2). It is necessary to attract top-level coordinators *for counteraction* to the harmful purposes (often they name as “ideology”: Communist, Nazism, Religious, etc.), too.

The goals at country level *can be completely replaced* for other purposes owing to activity of the coordinators at the higher level within 10 years. As a result, the Country's population will be “polarized” by the new goal, which may look *completely* different in compared to the pre-existing. There are many examples in history: Alexander of Macedon, Napoleon, Bismarck, Hitler, Stalin and others. This is important for sustainable development and National security.

In detail will be described in separate paper. □

Example 9. Democratic mechanisms are not applicable to the coordinators.

Democratic mechanisms can be used to make decisions only within the level in the pyramid of behavior. The coordinator's decision is always a crucial for replicators: coordinator *always* “wins” by *any number* of replicators.

As a result, decision-making institutions at the level of large firms and corporations, countries and humanity as a whole should be changed. Today there is a high level of risk that the behavior of the corporation, country or humanity as a whole can be *ineffective*. With the development of civilization, this risk will grow rapidly.

In particular, democratic mechanisms are *an impediment* to economic growth in countries, in which the conditions for the formation and decay of *all* the pyramids of behavior have not created. The countries in former Soviet Union are examples: the legislation is democratic, but economic development is absent.

In developed countries the conditions for the formation of new and the destruction of inefficient pyramids of behavior are only partially created in economics. For example, in the areas of finance, such conditions do not exist. Moreover, the necessity of these conditions is not even recognized: this is demonstrated by the behavior of the U.S. government *after* 2008.

In detail will be described in separate paper. □

Example 10. The P versus NP Problem solving in the frame of 2AIO.

A detailed description of the Problem sees in (Cook, 2011). Class P problems can be defined as a class chain of decisions, which are *deterministic* and have polynomial length and polynomial-time solution. While the class of NP problems is *indeterminist* and has a length of time and solve more than a polynomial.

In frame of the 2AIO instrument this problem can be formulated as follows. There are two classes of problems for the hierarchical level of the pyramid of behavior. The first class of problems is those that can be solved by replicators at the hierarchical level. This requires a chain of 2AIO by *finite* length (which depends on the type 2AO and interior, in which such 2AIO do the behavior). The number of types 2AIO in such chains is *finite* and depends on the type of problem.

The second class of problems is those that can only be resolved by the coordinators for this hierarchical level. When a coordinator is absent, then *any number of replicators* will not solve the problem. The sole algorithm, which available to replicators, is a sort out method or the method of trial and error (and it requires a non-polynomial time).

We can say that the pyramid of behavior is just created in order to solve the class NP problems in polynomial time. It is interesting that there is a clear correspondence with the theorems of Gödel's proof of the consistency of the axiom system: optimal behavior for a given level of the hierarchical pyramid has to offer (to "prove"), only the coordinator. □

Conclusions

Constructed instrument is powerful for solving a number of key problems of economics. With it using the description, forecast and construction of optimal technologies can be constructed for both the behavior of the person and cooperative behavior of aggregates of people (from small businesses to government and mankind in general).

However, this instrument proved to be quite easy to use.

The obtained results require significant changes to existing institutions.

Further development of mankind will be significantly delayed in the case, when the instruments, which are described in the paper, will not be used.

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